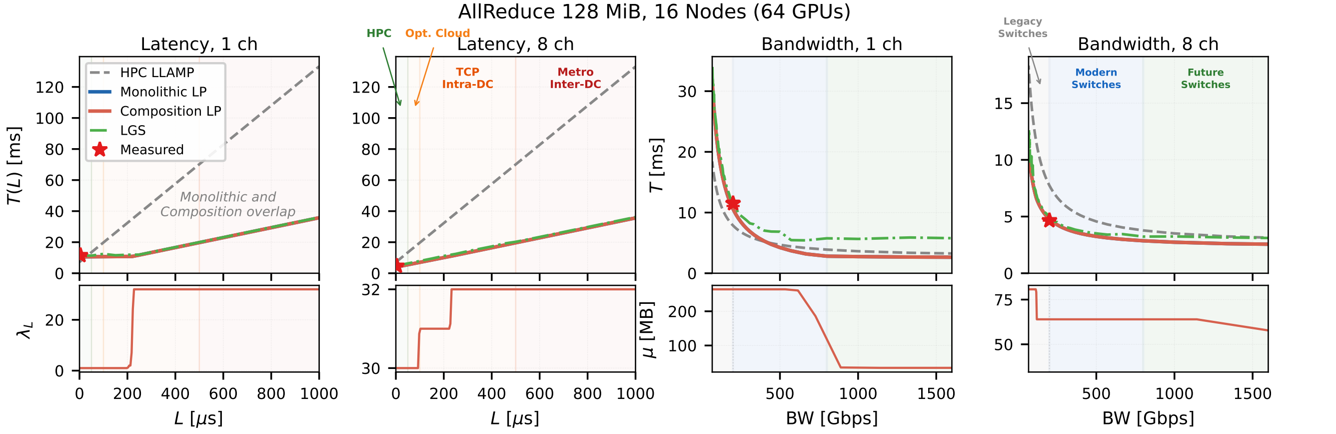
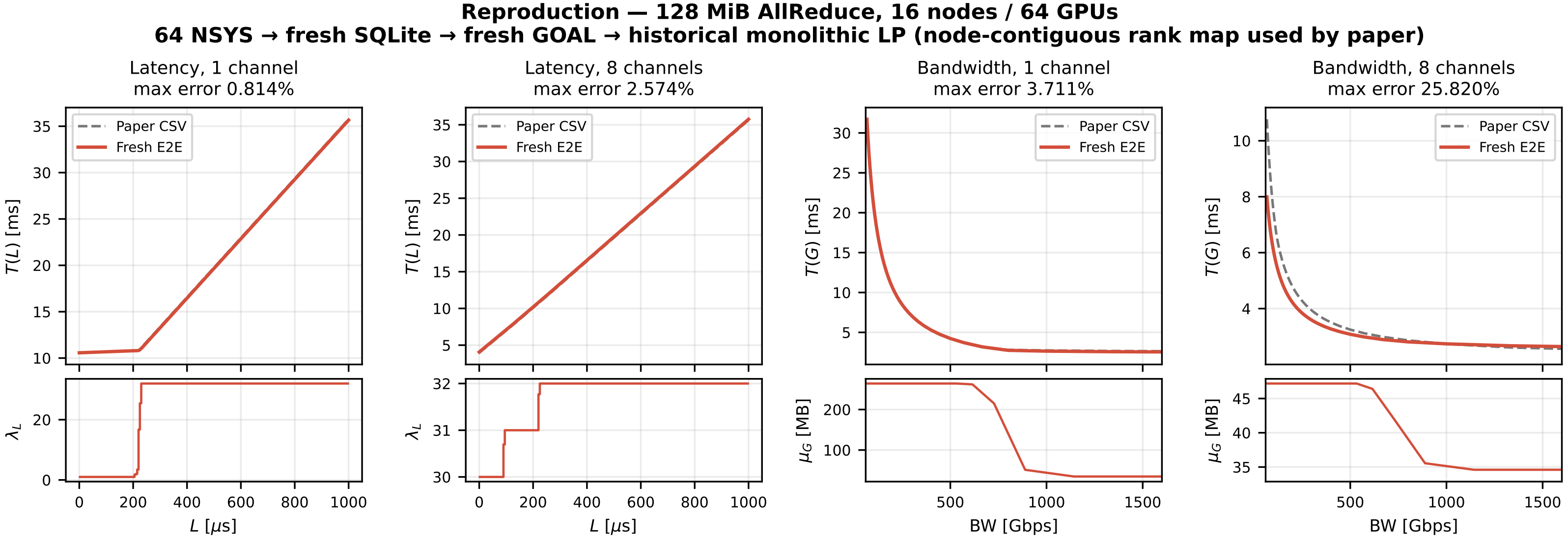
Figure 3 — AllReduce 128 MiB, 16 nodes / 64 GPUs

PAPER RESULT (verbatim crop from the supplied PDF)



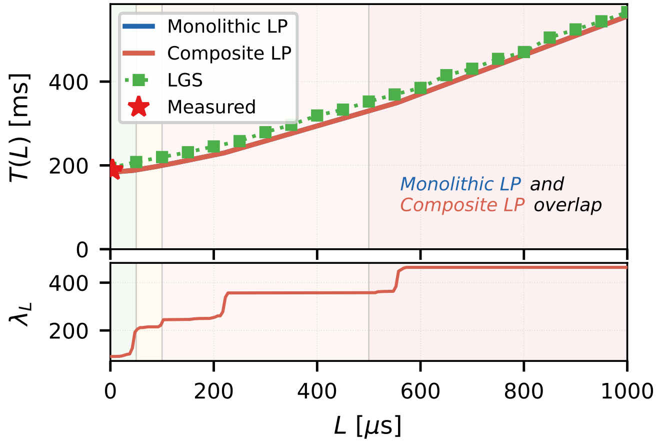
REPRODUCED RESULT — NSYS → fresh SQLite → fresh GOAL → historical monolithic LP → plot



Best paper-era configuration uses node-contiguous rank blocks. Max error: latency 0.814% / 2.574%; bandwidth 3.711% / 25.820% (1ch / auto). Auto-bandwidth remains a compositional-model gap.

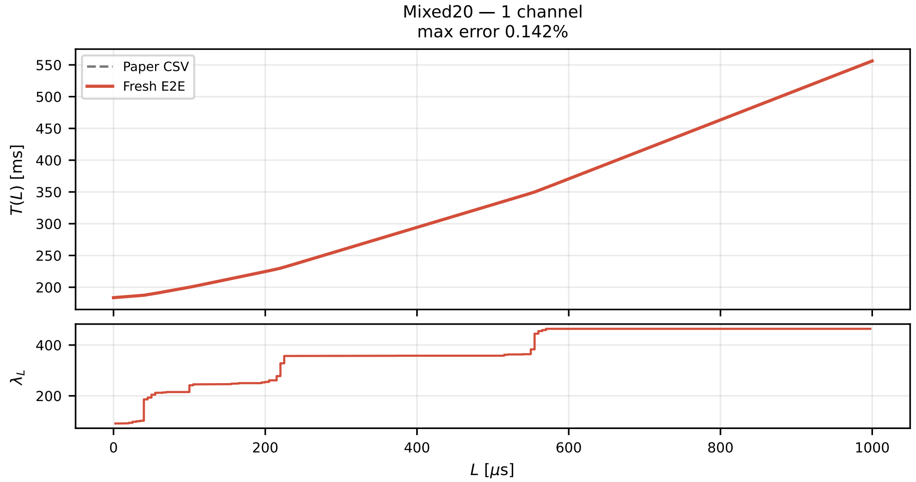
Figure 4 — Mixed20, 16 nodes / 64 GPUs, forced one channel

PAPER RESULT (verbatim crop from the supplied PDF)



REPRODUCED RESULT — 64 NSYS → fresh SQLite → 67.4 MB GOAL → barrier-aware monolithic LP → plot

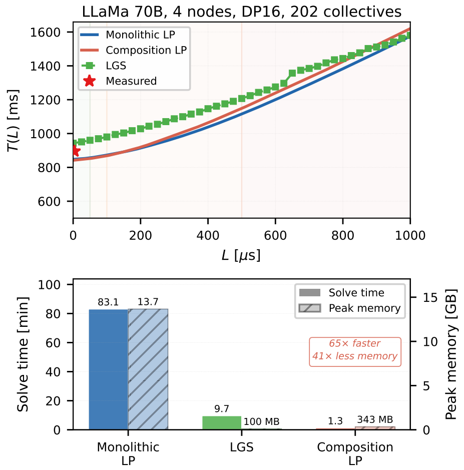
Reproduction — 20 collectives, 16–64 MiB, 16 nodes / 64 GPUs
64 NSYS → fresh SQLite → 67.4 MB GOAL → 617,607-variable LP + barriers | independent BW sweep max error 0.242%



Near-exact: latency max/mean error 0.142% / 0.047%. The independent bandwidth sweep also matches to 0.242% max error. This is the older Mixed20 campaign used by the plotted paper curve, not corrected job 1791883.

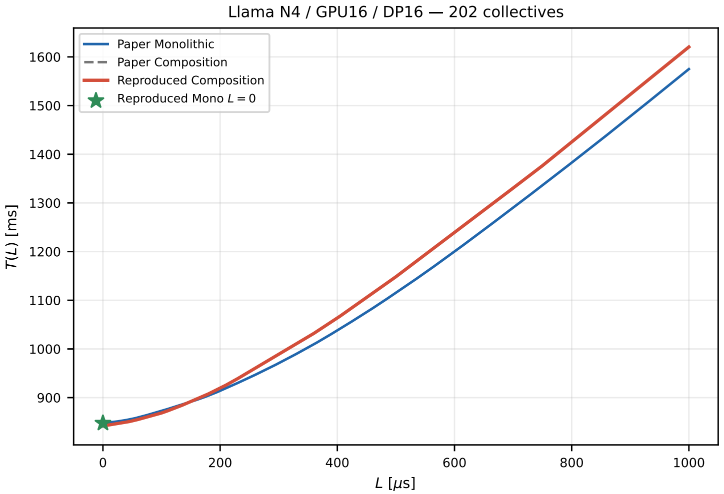
Figure 5 — Llama N4 / GPU16 / DP16

PAPER RESULT (verbatim crop from the supplied PDF)



REPRODUCED RESULT — public NSYS → fresh metadata → preserved April-5 motif LPs → composition → plot

Reproduction — trace-derived metadata to plot (preserved historical solver)



Pipeline provenance

4 public NSYS reports
↓ fresh SQLite / NCCL metadata
April-5 per-motif LP solves
↓ node-map + NIC compatibility
Sequential composition
↓
201-point sensitivity curve

Composite max error: 0.01485%
Composite mean error: 0.00488%
Monolithic $L = 0$ error: 0.000265%
Full Monolithic $L = 1$ ms: timed out
after 1,345 s (5.0M constraints)

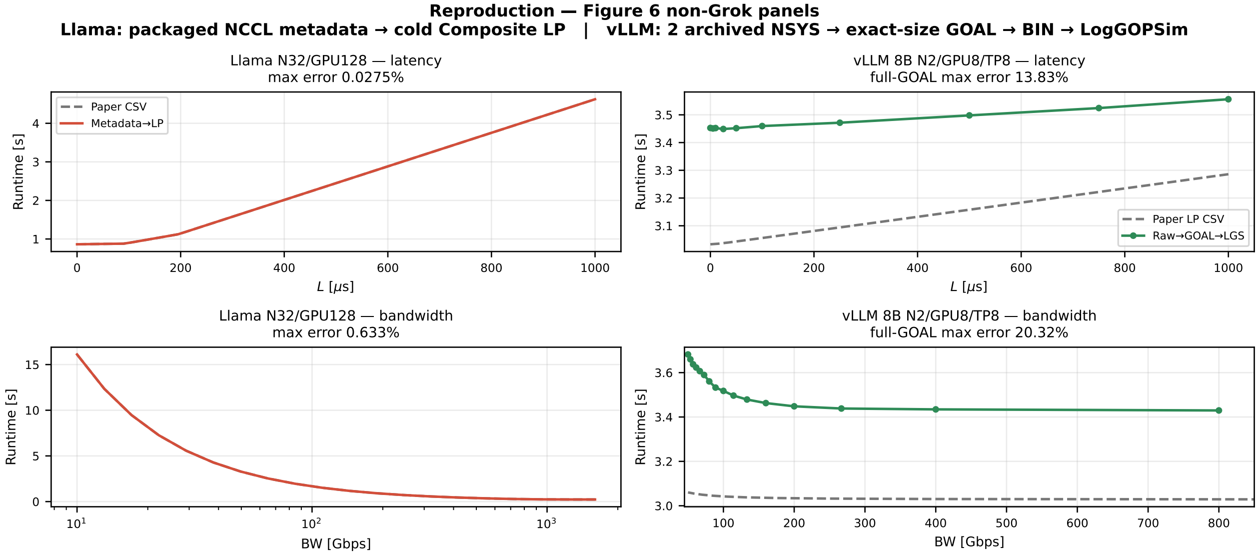
Observed model is Llama 7B/6.74B.
The paper's 70B label is treated as a typo.

Observed trace is Llama 7B/6.74B; the paper says 70B and is treated as a label typo. Composite max error is 0.01485%; Monolithic $L = 0$ error is 0.000265%. The 5.0M-constraint Monolithic $L = 1$ ms solve hit its 1,345 s time limit.

Figure 6 (non-Grok) — Llama training and vLLM inference

PAPER RESULT (left and right columns only; Grok deliberately excluded)

REPRODUCED RESULT — Llama: metadata → cold Composite LP; vLLM: 2 exact NSYS → exact-size GOAL → BIN → LGS



Llama (observed 7B, paper says 70B): max error 0.0275% latency / 0.633% bandwidth. vLLM raw chain is proven and exactly reproduces job 1812658 at $L = 3.7\,\mu s$, but the paper LP CSV is 13.83–20.32% lower; its unrecovered 8.5 MB crop/compression step is still missing.